

Spatial Filters

Sharpening Spatial Filters

The principal objective of sharpening is to highlight transitions of intensity.

Transition intensity: to change (intensity) from an origin state to a destination state at time t .

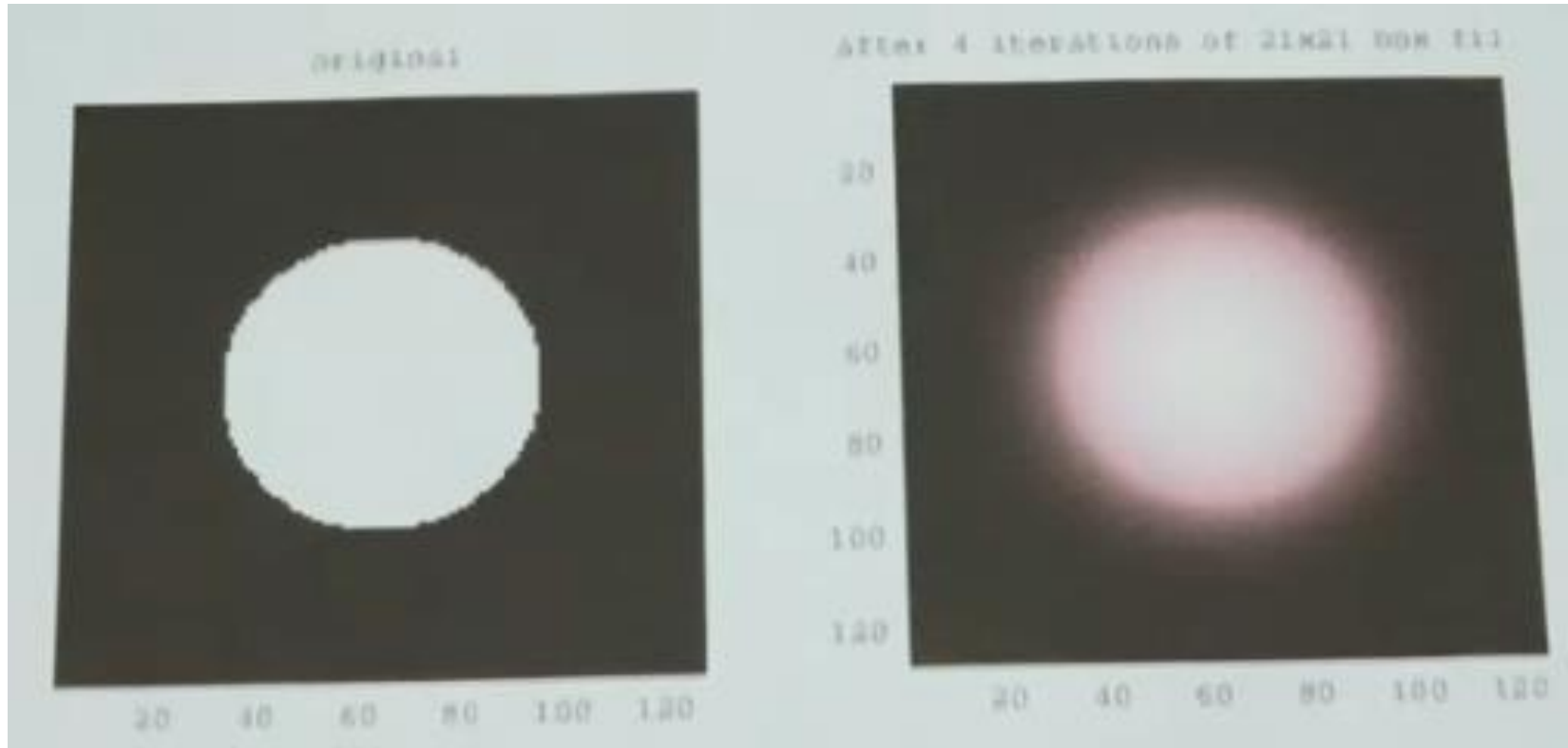
Applications of image sharpening include electronic printing, medical imaging, industrial inspection and autonomous guidance in military systems.

- Blurring – Pixel averaging
- Sharpening – spatial differentiation

The strength of the response of a derivational operator is proportional to the degree of intensity discontinuity of the image at the point at which the operator is applied.

Therefore, image differentiation enhances edges and other discontinuities (such as noise) and deemphasizes area which have slowly varying intensities.

Sharpening Spatial Filters

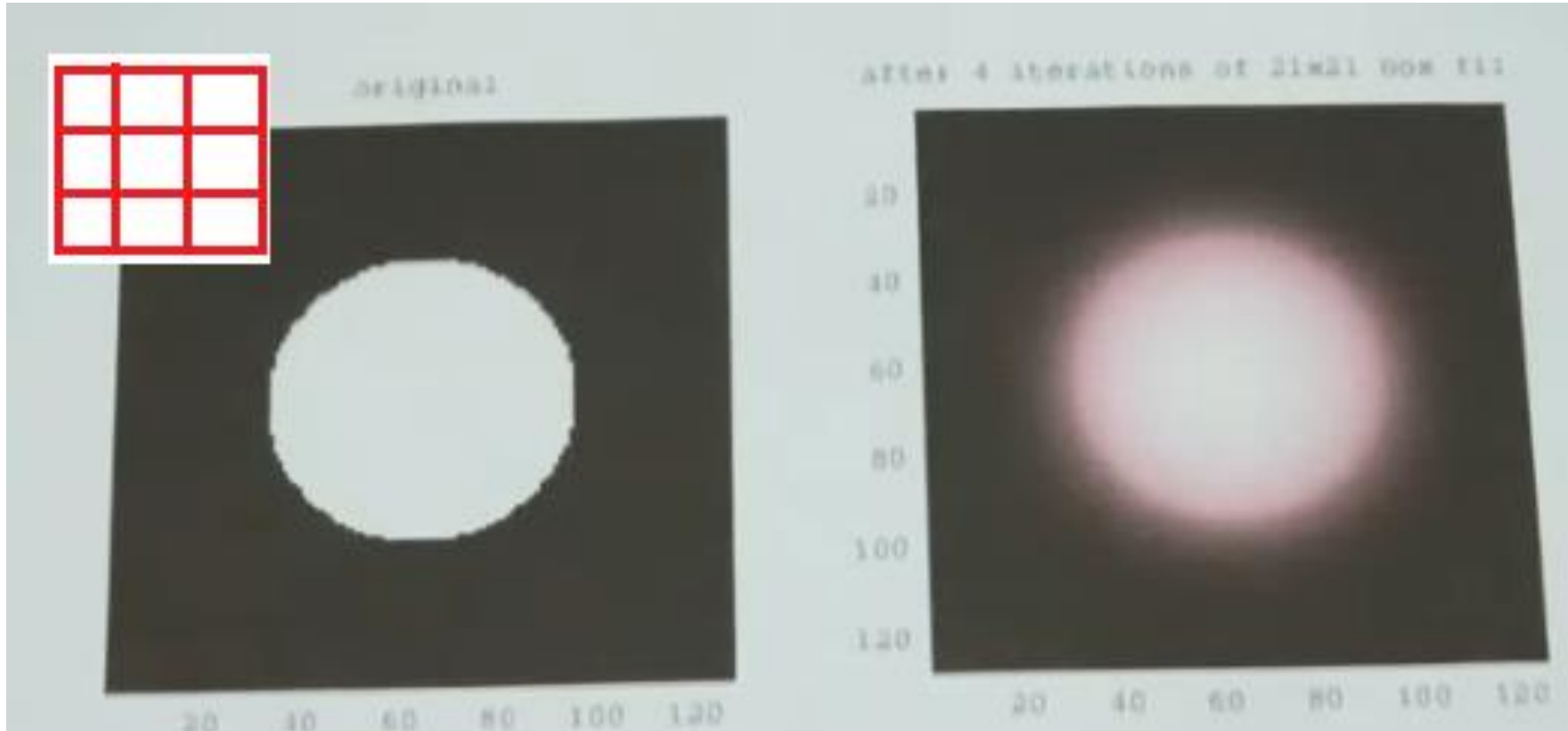


Left – original Image

Right – Blurred Image

There will be a mask which will slide the original image and will result the blurred image (average value will be placed in the center pixel to obtain the result)

Sharpening Spatial Filters



Left – original Image

Right – Blurred Image

There will be a mask which will slide the original image and will result the blurred image (average value will be placed in the center pixel to obtain the result)

After 4 iterations of 21 x 21 box filter the result will be right one

Foundation of sharpening filters

1) First-order derivatives of a one-dimensional function

$f(x)$:

$$\frac{\partial f}{\partial x} = f(x + 1) - f(x)$$

2) Second-order derivative of a one-dimensional function

$f(x)$:

$$\frac{\partial^2 f}{\partial x^2} = f(x + 1) + f(x - 1) - 2f(x)$$

Laplacian Filter

- It highlights gray-level discontinuities in an image
- It deemphasizes regions with slowly varying gray levels

Formula:

$f(x)$:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Where, $\frac{\partial^2 f}{\partial x^2} = f(x + 1, y) + f(x - 1, y) - 2f(x, y)$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y + 1) + f(x, y - 1) - 2f(x, y)$$

Previously we had 1 dimensional now we will consider 2 dimensional

Laplacian Filter

$$\begin{aligned}\nabla^2 f &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \\ &= [f(x+1, y) + f(x-1, y) - 2f(x, y)] + \\ &\quad [f(x, y+1) + f(x, y-1) - 2f(x, y)] \\ &= f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)\end{aligned}$$

Laplacian Filter

0	1	0
1	-4	1
0	1	0

1	1	1
1	-8	1
1	1	1

$f(x-1,y-1)$	$f(x,y-1)$	$f(x+1,y-1)$
$f(x-1,y)$	$f(x,y)$	$f(x+1,y)$
$f(x-1,y+1)$	$f(x,y+1)$	$f(x+1,y+1)$

0	-1	0
-1	4	-1
0	-1	0

-1	-1	-1
-1	8	-1
-1	-1	-1

Laplacian Filter

8	5	4
0	6	2
1	3	7

Input Image

0	1	0
1	-4	1
0	1	0

Mask

$$\begin{aligned} &= (8 \times 0) + (5 \times 1) + (4 \times 0) + (0 \times 1) + (6 \times -4) + (2 \times 1) + (1 \times 0) + (3 \times 1) + (7 \times 0) \\ &= 0 + 5 + 0 + 0 - 24 + 2 + 0 + 3 + 0 \\ &= -14 \end{aligned}$$

Enhanced Laplacian Filter

Enhanced Laplacian Filter

Example 1:

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow{\text{Enhanced}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & -5 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Example 2:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{\text{Enhanced}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & -9 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Q2: Apply enhanced laplacian filter on the given image on the center pixel.

8	5	4
0	6	2
1	3	7

Rule: will increase 1 in the center point (e.g.; if -4 then it will be -5, if +4 then it will be +5)

Enhanced Laplacian Filter

Enhanced Laplacian Filter

0 1 0

1 -4 1

0 1 0

Enhanced

→

0 1 0

1 -5 1

0 1 0

1 1 1

1 -8 1

1 1 1

Enhanced

→

1 1 1

1 -9 1

1 1 1

Q2. Apply enhanced laplacian filter on the given image on the center pixel.

8	5	4
0	6	2
1	3	7

*

1 1 1

1 -9 1

1 1 1

Enhanced Laplacian Filter

Enhanced Laplacian Filter

0 1 0

1 -4 1

0 1 0

Enhanced

→

0 1 0

1 -5 1

0 1 0

1 1 1

1 -8 1

1 1 1

Enhanced

→

1 1 1

1 -9 1

1 1 1

Q2. Apply enhanced laplacian filter on the given image on the center pixel.

8	5	4
0	6	2
1	3	7

*

1	1	1
1	-9	1
1	1	1

$$= 8 + 5 + 4 + 0 - 54$$

$$+ 2 + 1 + 3 + 7$$

$$= 30 - 54 = \underline{\underline{-24}}$$

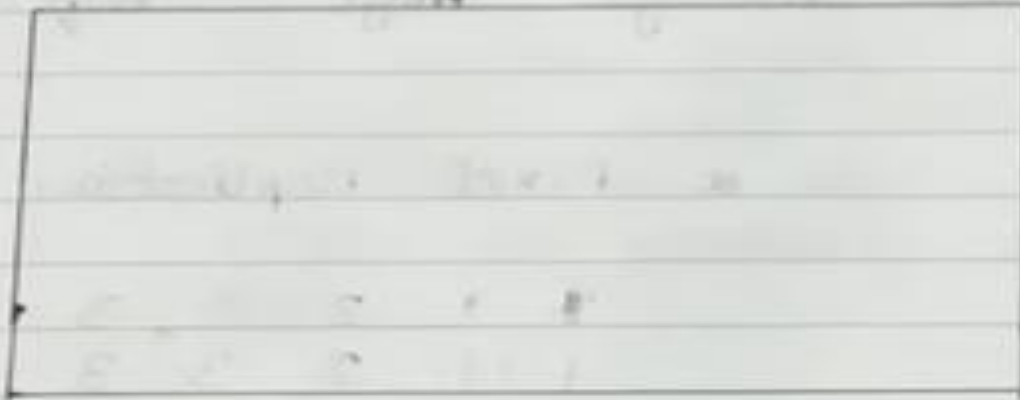
Enhanced Laplacian Filter

Q3. Apply Laplacian filter on the given image.

50	50	50	100	100	100
50	50	50	100	100	100
50	50	50	100	100	100
100	100	100	50	50	50
100	100	100	50	50	50
100	100	100	50	50	50

1	1	1
1	-8	1
1	1	1

Mask



Enhanced Laplacian Filter

Q 3. Apply Laplacian filter on the given image.

50	50	50	50	100	100	100	100
50	50	50	50	100	100	100	100
50	50	50	50	100	100	100	100
50	50	50	50	100	100	100	100
100	100	100	100	50	50	50	50
100	100	100	100	50	50	50	50
100	100	100	100	50	50	50	50
100	100	100	100	50	50	50	50

1	1	1
1	-8	1
1	1	1

Mask

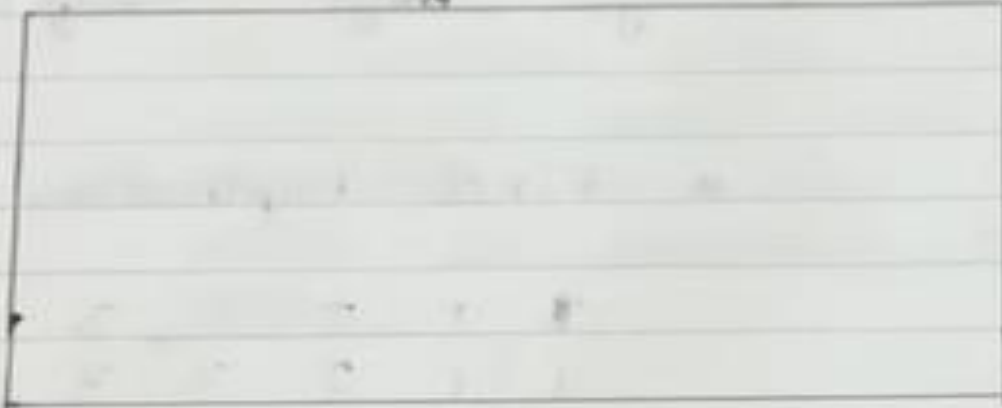
Enhanced Laplacian Filter

Q 3. Apply Laplacian filter on the given image.

50	50	50	50	100	100	100	100
50	50	50	50	100	100	100	100
50	50	50	50	100	100	100	100
50	50	50	50	100	100	100	100
100	100	100	100	50	50	50	50
100	100	100	100	50	50	50	50
100	100	100	100	50	50	50	50
100	100	100	100	50	50	50	50

1	1	1
1	-8	1
1	1	1

Mask



Enhanced Laplacian Filter

Q 3. Apply Laplacian filter on the given image.

50	50	50	50	100	100	100	100
50	50	50	50	100	100	100	100
50	50	50	50	100	100	100	100
50	50	50	50	100	100	100	100
100	100	100	100	50	50	50	50
100	100	100	100	50	50	50	50
100	100	100	100	50	50	50	50
100	100	100	100	50	50	50	50

1	1	1
1	-8	1
1	1	1

Mask

$$50 \times 8 - 50 \times 8 \\ = 0$$

Enhanced Laplacian Filter

Q 3. Apply Laplacian filter on the given image.

50	50	50	50	100	100	100	100
50	50	50	50	100	100	100	100
50	50	50	50	100	100	100	100
50	50	50	50	100	100	100	100
100	100	100	100	50	50	50	50
100	100	100	100	50	50	50	50
100	100	100	100	50	50	50	50
100	100	100	100	50	50	50	50

1	1	1
1	-8	1
1	1	1

Mask

$$50 \times 8 - 50 \times 8$$

$$= 0$$

$$50 \times 5 + 100 \times 3 - 50 \times 8$$

$$= 450 + 300 - 400$$

$$= 150$$

$$50 \times 3 + 100 \times 5 - 400$$

$$= 150 + 500 - 400$$

$$= -150$$

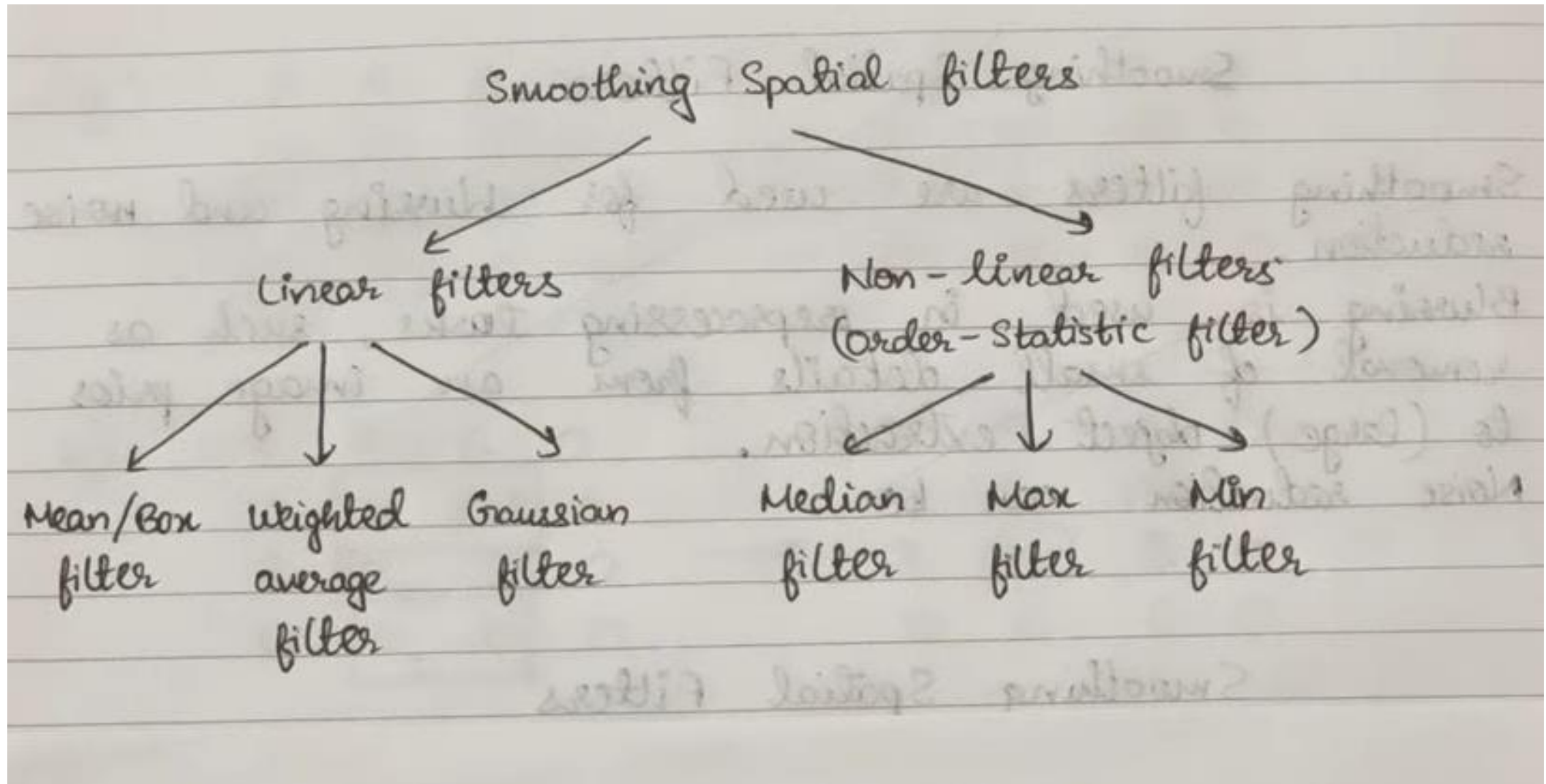
0	0	150	-150	0	0
0	0	150	-150	0	0
150	150	200	-200	-150	-150
-150	-150	-200	200	150	150
0	0	-150	150	0	0
0	0	-150	150	0	0

Smoothing Spatial Filter

Smoothing Spatial Filters

- Smoothing filters are used for blurring and noise reduction.
- Blurring is used in preprocessing tasks, such as removal of small details from an image prior to (large) object extraction.
- Noise reduction can be accomplished by blurring with a linear filter and also by non-linear filtering.

Smoothing Spatial Filter



Smoothing Spatial Filter

* Smoothing linear filters

They are also known as averaging filters (or) lowpass filters as they are simply the average of the pixels contained in the neighbourhood of the filter mask.

The process results in an image with reduced 'sharp' transitions in intensities which ultimately leads to noise reduction.

Smoothing Spatial Filter

1) Box filter - all coefficients are equal.

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \rightarrow \text{Mask}$$

2) Weighted average - give more (less) weight to pixels near (away from) the output location.

$$\frac{1}{16} \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \rightarrow \text{Mask}$$

Smoothing Spatial Filter

3) Gaussian filter - the weights are samples of 2D Gaussian function:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$

(2D Gaussian function)
 $\sigma \rightarrow$ Standard deviation

$$\frac{1}{16} \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \rightarrow \text{Mask}$$

- $\rightarrow$ Used to blur edges and reduce contrast.
- $\rightarrow$ Similar to median filter but is faster.

Smoothing Spatial Filter

* Non-linear (Order-Statistic) filters

Their response is based on ordering (ranking) the pixels contained in the image area encompassed by the filter, and then replacing the value of the center pixel with the value determined by the ranking result.

- 1) Median filter - Find the median of all the pixel values.
- 2) Min filter - Find the minimum of all the pixel values.
- 3) Max filter - Find the maximum of all the pixel values.

Smoothing Spatial Filter

Q1. Consider the image below and calculate the output of the pixel (2,2) if smoothing is done using 3×3 neighbourhood using all the filters below:

- a) Box / Mean filter
- b) Weighted average filter
- c) Median filter
- d) Min filter
- e) Max filter

1	8	8	0	7
4	7	9	5	7
5	4	6	8	6
4	2	0	1	5
0	1	0	2	0

3×3

Smoothing Spatial Filter

Q1. Consider the image below and calculate the output of the pixel (2,2) if smoothing is done using 3×3 neighbourhood using all the filters below:

- a) Box / Mean filter
- b) Weighted average filter
- c) Median filter
- d) Min filter
- e) Max filter

1	8	8	0	7
4	7	9	5	7
5	4	6	8	6
4	2	0	1	5
0	1	0	2	0

a) Box filter

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Smoothing Spatial Filter

Q1. Consider the image below and calculate the output of the pixel (2,2) if smoothing is done using 3×3 neighbourhood using all the filters below:

- a) Box / Mean filter
- b) Weighted average filter
- c) Median filter
- d) Min filter
- e) Max filter

1	8	8	0	7
4	7	9	5	7
5	4	6	8	6
4	2	0	1	5
0	1	0	2	0

3×3

a) Box filter

$$= \frac{1}{9} \times [7 + 9 + 5 + 4 + 6 + 8 + 2 + 0 + 1] \times \frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \frac{1}{9} \times [42] = 4.66 \approx \underline{\underline{5}}$$

Smoothing Spatial Filter

Consider the image below and calculate the output of the pixel (2,2) if smoothing is done using 3×3 neighbourhood using all the filters below:

- a) Box / Mean filter
- b) Weighted average filter
- c) Median filter
- d) Min filter
- e) Max filter

1	8	8	0	7
4	7	9	5	7
5	4	6	8	6
4	2	0	1	5
0	1	0	2	0

b) Weighted average filter

$$= \frac{1}{16} \left[\begin{array}{l} 7 \times 1 + 9 \times 2 + 5 \times 1 + 4 \times 2 + \\ 4 \times 6 + 8 \times 2 + 2 \times 1 + 0 \times 2 \\ + 1 \times 1 \end{array} \right]$$

$$= \frac{1}{16} [81] = 5.0625 \approx \underline{\underline{5}}$$

$$\frac{1}{16} \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Smoothing Spatial Filter

Consider the image below and calculate the output of the pixel (2,2) if smoothing is done using 3×3 neighbourhood using all the filters below:

- a) Box / Mean filter
- b) Weighted average filter
- c) Median filter
- d) Min filter
- e) Max filter

1	8	8	0	7
4	7	9	5	7
5	4	6	8	6
4	2	0	1	5
0	1	0	2	0

- c) Median filter
0, 1, 2, 4, (5), 6, 7, 8, 9

Median = 5

d) Min filter
= 0

e) Max filter
= 9

Smoothing Spatial Filter

First Order Derivative Filters

1) Roberts operators (Cross-gradient)

-1	0
0	1

0	-1
1	0

z1	z2	z3
z4	z5	z6
z7	z8	z9

Image

2) Sobel operators

-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

3) Prewitt operators

-1	-1	-1
0	0	0
1	1	1

-1	0	1
-1	0	1
-1	0	1

Mainly used for edge detection

Smoothing Spatial Filter

Problems with Roberts Cross :

- 1) 2×2 masks are not easy to implement.
- 2) No. of calculations are more.
- 3) No. of neighboring pixels considered in one go are less.

To solve these problems, we make the following changes:

- 1) Change in the size of the mask.
- 2) Change in the no. of neighbouring pixels considered.

Smoothing Spatial Filter

Q2. Apply Roberts, Sobel and Prewitt operators on the pixel (1,1) in the following image.

50	50	100	100
50	50	100	100
50	50	100	100
50	50	100	100

Smoothing Spatial Filter

Q2. Apply Roberts, Sobel and Prewitt operators on the pixel (1,1) in the following image.

50	50	100	100
50	50	100	100
50	50	100	100
50	50	100	100

Input image

1) Roberts operator

-1	0
0	1

Smoothing Spatial Filter

Q2. Apply Roberts, Sobel and Prewitt operators on the pixel (1,1) in the following image.

50	50	100	100
50	50	100	100
50	50	100	100
50	50	100	100

Input image

1) Roberts operator

-1	0
0	1

$$\begin{aligned} &= 50 \times (-1) + 100 \times 1 \\ &= -50 + 100 = \underline{\underline{50}} \end{aligned}$$

Smoothing Spatial Filter

Q2. Apply Roberts, Sobel and Prewitt operators on the pixel (1,1) in the following image.

50	50	100	100
50	50	100	100
50	50	100	100
50	50	100	100

Input image

1) Roberts operator

-1	0
0	1

$$\begin{aligned} &= 50 \times (-1) + 100 \times 1 \\ &= -50 + 100 = \underline{\underline{50}} \end{aligned}$$

2) Sobel operator

-1	-2	-1
0	0	0
1	2	1

$$\begin{aligned} &= 50(-1) + 50(-2) + 100(-1) + 50(1) + 50(2) + 100(1) \\ &= -50 - 100 - 100 + 50 + 100 + 100 = \underline{\underline{0}} \end{aligned}$$

Smoothing Spatial Filter

3) Prewitt operator

$$\begin{bmatrix} 50 & 50 & 100 & 100 \\ 50 & \textcircled{50} & 100 & 100 \\ 50 & 50 & 100 & 100 \\ 50 & 50 & 100 & 100 \end{bmatrix} * \begin{bmatrix} -1 & -1 & -1 \\ 0 & \textcircled{0} & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Input image

$$\begin{aligned} &= -1(50 + 50 + 100) + 1(50 + 50 + 100) \\ &= \underline{\underline{0}} \end{aligned}$$

Thanks !!!